Let $\vec{u}$ be the position vector of the meeting point of the line with the plane. Then $\vec{u}$ satisfies both $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} \cdot \vec{n} = p$ for some value of t, say t_1. So, We get

$\vec{u} = \vec{a} + t\vec{b}$ ... (1)

$\vec{u} \cdot \vec{n} = p$ ...(2)

Sustituting (1) in (2), we get

$$\left(\vec{a} + t_1\vec{b}\right) \cdot \vec{n} = p$$

or $$\vec{a} \cdot \vec{n} + t_1\left(\vec{b} \cdot \vec{n}\right) = p$$

or $$t_1 = \frac{p - \left(\vec{a} \cdot \vec{n}\right)}{\vec{b} \cdot \vec{n}} \quad \text{...(3)}$$

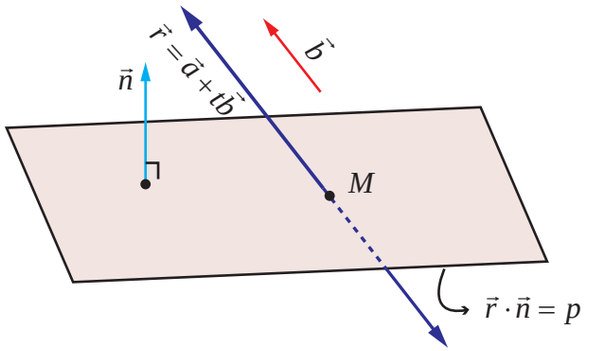


Fig. 6.36

Sustituting (3) in (1), we get

$$\vec{u} = \vec{a} + \left(\frac{p - \left(\vec{a} \cdot \vec{n}\right)}{\vec{b} \cdot \vec{n}}\right)\vec{b},\ \vec{b} \cdot \vec{n} \neq 0$$

Example 6.56

Find the coordinates of the point where the straight line $\vec{r} = \left(2\hat{i} - \hat{j} + 2\hat{k}\right) + t\left(3\hat{i} + 4\hat{j} + 2\hat{k}\right)$ intersects the plane $x - y + z - 5 = 0$.

Solution

Here, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k},\ \vec{b} = 3\hat{i} + 4\hat{j} + 2\hat{k}$.

The vector form of the given plane is $\vec{r} \cdot \left(\hat{i} - \hat{j} + \hat{k}\right) = 5$. Then $\vec{n} = \hat{i} - \hat{j} + \hat{k}$ and $p = 5$.

We know that the position vector of the point of intersection of the line $\vec{r} = \vec{a} + t\vec{b}$ and the plane $\vec{r} \cdot \vec{d} = p$ is given by $\vec{u} = \vec{a} + \left(\frac{p - \left(\vec{a} \cdot \vec{n}\right)}{\vec{b} \cdot \vec{n}}\right)\vec{b}$, where $\vec{b} \cdot \vec{n} \neq 0$.

Clearly, we observe that $\vec{b} \cdot \vec{n} \neq 0$.

Now, $\frac{p - \vec{a} \cdot \vec{n}}{\vec{b} \cdot \vec{n}} = \frac{5 - \left(2\hat{i} - \hat{j} + 2\hat{k}\right) \cdot \left(\hat{i} - \hat{j} + \hat{k}\right)}{\left(3\hat{i} + 4\hat{j} + 2\hat{k}\right) \cdot \left(\hat{i} - \hat{j} + \hat{k}\right)} = 0$. Therefore,the position vector of the point of intersection of the given line and the given plane is

$$\vec{r} = \left(2\hat{i} - \hat{j} + 2\hat{k}\right) + (0)\left(3\hat{i} + 4\hat{j} + 2\hat{k}\right) = 2\hat{i} - \hat{j} + 2\hat{k}$$

That is, the given straight line intersects the plane at the point $(2, -1, 2)$.

Aliter

The Cartesian equation of the given straight line is $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{2} = t$ (say)

We know that any point on the given straight line is of the form $(3t + 2, 4t - 1, 2t + 2)$. If the given line and the plane intersects, then this point lies on the given pane $x - y + z - 5 = 0$.

So, $(3t + 2) - (4t - 1) + (2t + 2) - 5 = 0 \Rightarrow t = 0$.

Therefore, the given line intersects the given plane at the point $(2, -1, 2)$

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